RootCause AI Backend Master Prompt - Part 2

Controllers:
- authController
- userController
- analysisController
- dashboardController

Services:
1. promptBuilderService
Combine bug report, logs, stack trace, code snippet and environment into one structured prompt.
Request ONLY JSON from Gemini.

2. geminiService
Read GEMINI_API_KEY from .env.
Call model gemini-2.5-flash.
Return parsed JSON.

3. analysisService
Validate uploads.
Extract text.
Call promptBuilder.
Call geminiService.
Save result to MongoDB.
Return analysisId.

Routes:
POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout
POST /api/auth/forgot-password

GET /api/user/profile
PUT /api/user/profile

POST /api/analysis/upload
GET /api/analysis
GET /api/analysis/:id
DELETE /api/analysis/:id

GET /api/dashboard/stats

Dashboard Stats Response:
{
totalAnalyses,
highSeverity,
mediumSeverity,
lowSeverity
}

Upload Pipeline:
Client Upload
-> Multer
-> Validate file type & size

-> Store uploads
-> Extract text
-> Build AI prompt
-> Gemini
-> JSON
-> MongoDB
-> Return analysisId

Middleware:
verifyJWT
uploadMiddleware
validationMiddleware
errorMiddleware
helmet
cors
rateLimiter
morgan

Validation:
Email
Password strength
Required fields
Allowed mime types
Max upload size

API Response Format:
Success:
{success:true,message:'',data:{}}

Failure:
{success:false,message:'',errors:[]}

Analysis Detail Response:
uploadedFiles
summary
reproductionSteps
rootCause
confidence
suggestedFiles
fixRecommendations
severity
priority
createdAt